

Ashutosh Joshi

Software Engineer — AI/ML Systems

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SUMMARY

Software engineer with 4+ years building data pipelines, scientific instrument software, and production AI/LLM systems. Currently engineering the data backbone of NASA's Pandora atmospheric network — 300+ spectrometers across five continents — turning raw sensor streams into NASA/ESA science. Ship end-to-end: C++17 processing engines, ML anomaly detection, RAG and multi-agent platforms, and full-stack products.

PROFESSIONAL EXPERIENCE

Software Engineer — SciGlob Instruments & Services (NASA GSFC — Pandora Global Network), Columbia, MD **Nov 2024 - Present**

- Engineered automated data acquisition and cleaning pipelines aggregating atmospheric measurements from **300+ globally deployed spectrometers**, delivering analysis-ready datasets to NASA/ESA researchers.
- Designed and deployed **ML/deep-learning anomaly detection** across the full instrument fleet, automatically flagging sensor and data irregularities and eliminating hours of daily manual monitoring.
- Built an **encoder-decoder CNN for cloud detection** and integrated it into the sun-scan workflow, gating measurements on live sky conditions to raise scan quality network-wide.
- Developed a fleet-health monitoring web dashboard (device health, data flow, logging, operational state) for all 300+ units, accelerating issue triage across five continents.
- Leading a ground-up **C++17 rewrite of the Blick spectral processing suite (L0→L2)**: 14 modular libraries, parity test harness against legacy output, and a GPU-ready acceleration framework.
- Engineered fault-tolerant hardware control software (motors, spectrometers, lasers, rain sensors) with safe command sequencing; shipped one-click Windows/macOS installers for lab and field systems.

Data Analyst & Developer — 407 Associates LLC, Laurel, MD

Apr 2024 - Nov 2024

- Built scalable Python ML pipelines (scikit-learn, PySpark, TensorFlow) powering predictive insights for product and operations teams.
- Managed AWS S3/Snowflake storage and complex SQL workloads; delivered Tableau and Power BI analytics that drove decisions **reducing operating costs by 25%**.

System Engineer — Tata Consultancy Services, Bangalore, IN

Jun 2020 - Aug 2022

- Engineered large-scale ETL pipelines for a major US grocery retailer: **10M+ rows** across sources with 20+ automated validation checks, cutting data-quality defects by **30%**.
- Improved cloud application reliability through performance testing and tuning: **+25% throughput**, **-30% response time**, sustaining 99.5-99.9% availability targets.
- Built deployment and monitoring automation with CI/CD across 3-5 environments, reducing release time by **40%** and incident MTTR by **25%**.

SELECTED PROJECTS

github.com/ashutoshjoshi1

- CommonGround — Enterprise RAG Platform**: multimodal document ingestion with OCR, semantic Q&A with citations, prompt versioning, evaluation datasets, and audit trails. *FastAPI · Next.js · pgvector · Redis · Docker*
- Claude TopstepX — LLM Futures Trading Engine**: LLM-driven decisions over a fully deterministic core — seven-dimension context scoring, risk-gate chain, immutable domain models, ~93% test coverage. *Python · Anthropic API · pytest*
- MoneyBall — Multi-Agent Sports Intelligence**: six specialized agents reason over stats, availability, news, venue, and matchups to synthesize NBA/IPL game predictions. *FastAPI · Next.js · PostgreSQL · Zep · OpenAI*
- ReBirth — AI Video Generation Platform**: audio synthesis, facial animation, refinement, and encoding orchestrated through a Celery job state machine with full observability. *Next.js · FastAPI · Celery · Redis · Docker*

TECHNICAL SKILLS

Languages: Python, C++17, TypeScript, SQL, Java

AI / ML: LLM systems, RAG, multi-agent architectures, evals, PyTorch, TensorFlow, scikit-learn, PySpark

Backend & Infra: FastAPI, Flask, Node.js, PostgreSQL/pgvector, Redis, Celery, Docker, CI/CD, AWS, GCP, Azure, Linux

Frontend: React, Next.js, React Native

EDUCATION

University of Maryland, Baltimore County — M.P.S., Data Science (ML concentration) · GPA 3.89/4.00

Aug 2022 - May 2024